Class Meeting Handout #15

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

We have seen a two-parameter plot in the Yukawa-Tsuno equation that separated inductive and resonance electronic effects of substituents. We will now explore a two-parameter system that separates electronic and steric effects as we consider the Taft equation. We will further define other common LFER schemes used for interpreting reaction mechanisms.[1]

[1] Read chapter 8.3, 8.4, 8.5 and 8.7.

. . . And the Rest

In the first season of Gilligan's Island (1964-1967),[2] the theme song listed the cast as Gilligan, the skipper, a millionaire, his wife and a movie star, but the professor and Mary-Anne were referred to only as "and the rest".[3] Quite often, some very useful LFER approaches are left for students to learn on their own. Today we will explore some of "the rest." There are many to choose from; let me know if I left out your favourite.

[2] This is actually the time period for many of the greatest advances in physical organic chemistry.

[3] For a the story of the theme song and how it was eventually changed to include everyone, see https://roomforall.com/outsights/and-the-rest/ (accessed on June 12, 2025) and listen to the original at https://www.youtube.com/watch?v=AMk2B89TxvY

Exercise: Other LFERs – 40 minutes

In your reading of the textbook you will have encountered the following LFER concepts. Answer the following questions to help yourself pay closer attention to them.

1. Write out the Grunwald-Winstein equation.[4] Define the terms and the reference reaction. How are Grunwald-Winstein solvent parameters, Y, useful? Why was the additional N_{OTs} term added to create the extended Grunwald-Winstein equation?

[4] See chapter 8.4.2 to 8.4.4

2. Write out the Swain-Scott equation.[5] Define the terms and the reference reaction. How are Swain-Scott nucleophilicity parameters, n_x, useful?

[5] See chapter 8.4.5

3. Explain how the Brønsted plot relates nucleophilicity and leaving group ability to the pK_a values of the conjugate acid of the nucleophile or leaving group.[6] How are Brønsted plots useful? What are the limitations of this approach?

[6] See chapter 8.5

4. Write out the Swain-Lupton equation.[7] Define the terms and the reference reaction. How are Swain-Lupton substituent parameters, F and R, useful? Can the Swain-Lupton substituent parameters be used to replace the set of σ, σ^+ and σ^- Hammett parameters?

[7] See chapter 8.3.5

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- be able to define and describe other LFER systems such as the Grunwald-Winstein equation, the Swain-Scott equation and the Brønsted plot.
- **Computer Skill:** be able to use an interactive *Python* notebook to interpret and analyze experimental data from the literature.
- **Computer Skill:** be able to use a text editor to create data sets using experimental data from the literature and make these data sets available for others.

A Look Ahead

Tools are needed to advance science. Often, the tools preceded the great discoveries and made them possible. Various LFER systems enable investigations of the nature of transition states and reaction mechanisms. More are developed every year to handle specific problems in chemistry. Each LFER system uses one or two changeable aspects in a reaction, develops a scoring system for these changes, and then relates how the changes affect a reference reaction to how they affect the reaction being investigated.

Ultimately, we hope to obtain a linear plot with a good correlation. But the world is changing. Big data is being cut down to size by machine learning tools. Imagine taking reaction data for every solvolysis reaction ever measured and allowing the AI system to determine its own parameters. Then we can use this AI model to interpret how the transition state changes with structure changes and changes in the surroundings. We won't be doing that, but you might do something like that in a few years.

In the next class, we will quickly survey other methods for elucidating reaction mechanisms. We can use product analysis, especially stereochemical analysis, to prove a hypothesis. We can interpret radical reactions using radical clocks. There are many other methods available. We will explore some ideas we might use later in this course when exploring literature examples of physical organic chemistry.